

Identification of Energy and Spatial Location of Electron Traps in AlGaIn/GaN HFET Structures

Shlomo Mehari, Arkady Gavrilov, Moshe Eizenberg, and Dan Ritter

Abstract—We present a methodology, based on gated van der Pauw measurements, to identify the electron trap charging or discharging energy at different spatial locations in AlGaIn/GaN heterostructures. The slow transient response of the 2-D electron gas concentration was recorded following different Schottky gate and substrate voltage pulses and at different temperatures. While recovering from a gate voltage stress, both trapping and detrapping processes are observed at different times and temperatures, demonstrating that dynamic rather than quasi-equilibrium considerations should be invoked to explain the data. Several distinct electron traps were identified, three traps located in the GaN buffer layer having detrapping activation energies of about 0.55, 0.45, and 0.21 eV, and an activation energy of about 0.64 eV was associated with trapping processes in the AlGaIn barrier layer.

Index Terms—Electron trapping effects, GaN, gated hall, heterostructure FETs (HFETs), high power, transient response.

I. INTRODUCTION

AlGaIn/GaN heterostructure FETs (HFETs) are promising devices for high power, large wideband, and high temperature applications [1]–[3] due to the unique combination of a wide energy bandgap, and a high 2-D electron gas (2-DEG) concentration. One of the key limiting factors of this technology is electron or hole trapping, causing threshold voltage instabilities in MIS HFET structures [4], [5], and current collapse or dynamic ON-resistance (R_{ON}) increase [6]–[10]. The GaN-based material system is notoriously known for containing a high density of defects [11]. In a Schottky gate HFET, the semiconductor electron traps are located in the AlGaIn barrier layer [12], [13] and in the GaN buffer layer [10], [14], both under the gate electrode or in the gate-drain access region. Additionally, a large trap density is located at the semiconductor surface in the access region [15], [16]. Extensive studies were carried out to assess the dynamic behavior of lateral AlGaIn/GaN HFETs, using primarily pulsed

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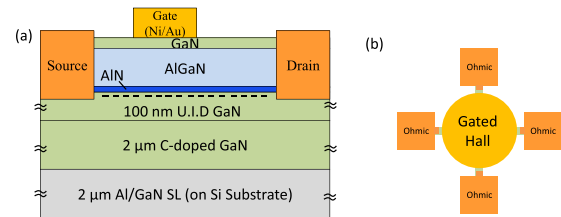


Fig. 1. (a) Schematic cross section of the HFETs layers on Si substrate. (b) Schematic top view of the gated van der Pauw test structures.

I – V techniques [6], [9], [17], [18], and switching transients techniques [19]. However, being inherently 2-D devices, the analysis of transistor transient behavior is complicated, and the determination of characteristic energy and spatial location of traps is not straightforward [10], [18], [20]. We recently introduced the gated van der Pauw method, which provides an opportunity to study electron trapping effects in the AlGaIn/GaN HFET structure [13], [21], as well as at the insulator/III–nitride interface [22]. Due to its 1-D nature, the analysis of the obtained results is more straightforward than the analysis of transistor pulsed I – V curves. In this paper, we present a methodology for the identification of electron trap characteristic energies at different spatial locations in GaN-based heterostructures. We emphasize that since charge redistribution involves transport processes, which are temperature-dependent, the extracted activation energies should be interpreted with care [13], [23].

The methodology described in this paper can be integrated into standard qualification measurements to assess the quality of the epitaxial layers and possibly also the fabrication process.

II. EXPERIMENTAL PROCEDURE

Fig. 1 shows a schematic cross-section of the HFETs structure and top view of the gated van der Pauw test structures. The semiconductor layers, provided by Nippon Telegraph and Telephone Corporation (Today NTT Advanced Technology Corporation), were grown using a metal organic chemical vapor deposition system on a (111) p-type Si substrate. The epilayers consist of, from the bottom, a 200-nm-thick AlN nucleation layer, a 2- μ m-thick AlN/GaN superlattice, a 2- μ m-thick carbon doped GaN layer, a 100-nm-thick unintentionally doped GaN channel layer, a 1-nm AlN layer followed by a 25-nm $\text{Al}_{0.25}\text{Ga}_{0.75}\text{N}$ barrier, and a 3-nm GaN cap layer. The 2-DEG channel is located at the AlN/GaN interface.

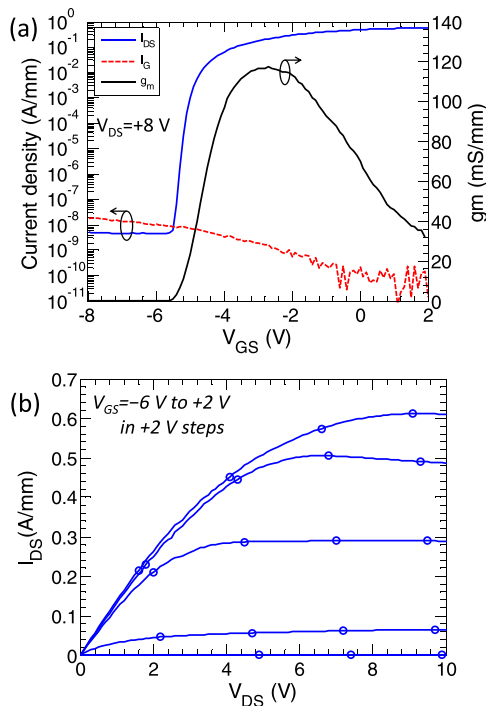


Fig. 2. (a) Semilog scale transfer characteristics. (b) Output curves of the HFETs.

Device fabrication started with e-beam evaporation of standard Ti/Al/Ni/Au (20/120/40/50 nm) ohmic contacts rapid thermal annealed at 825 °C for 1 min in N_2 atmosphere. Mesa isolation was then performed by reactive ion etching. A Ni/Au (30/200 nm) Schottky gate stack was e-beam evaporated without postdeposition treatments. The transistor gate dimensions were $2.5 \mu\text{m} \times 50 \mu\text{m}$ ($L_g \times W_g$), and the van der Pauw circular gate diameter was $100 \mu\text{m}$. The gate-to-drain spacing (L_{gd}) and the gate-to-source spacing (L_{gs}) were 4 and $2 \mu\text{m}$, respectively. Ti/Au (20/300 nm) metal pads were patterned on top of a polyamide passivation. For substrate back gating, a 200-nm-thick Al layer was e-beam evaporated on the back side of the p-type Si substrate. Conductivity measurements were performed by applying a constant current of $100 \mu\text{A}$ between two adjacent ohmic contacts and continuously measuring the voltage at the remaining two ohmic contacts. Hall measurements were performed at a magnetic field of 9000 G. Gate voltage was applied between the gate electrode and one of the four van der Pauw contacts.

III. RESULTS AND DISCUSSION

Typical dc characteristics of the fabricated AlGaIn/GaN HFETs on Si substrate are presented in Fig. 2. The transistors exhibited the state-of-the-art dc performance, with maximum drain current of about 0.6 A/mm at $V_{GS} = +2$ V, low OFF-state leakage current of about 2×10^{-8} A/mm, peak extrinsic transconductance of about 120 mS/mm at $V_{DS} = +8$ V, and low subthreshold slope of about 78 mV/decade.

We next briefly review the gated van der Pauw experiment. In case of a Schottky gate, surface traps are inactive, so that only barrier and buffer traps are probed [13], [21]. In the experiment, the 2-DEG concentration is monitored after a gate

TABLE I
SUMMARY OF APPLIED STRESS AND RELATED
PROBED TRAP LOCATION

Stress	Probed Traps	Comment
+3V top gate	Barrier traps only	Buffer traps screened by 2DEG
-3V top gate	Barrier and some buffer traps	Semi-on during stress
-10V top gate	Mostly buffer traps	Deep buffer depletion
+10V back gate	Buffer traps near substrate	

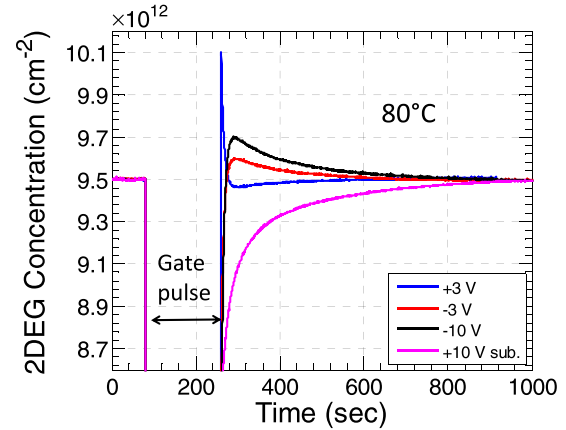


Fig. 3. 2-DEG transients after a different gate and substrate stress at 80 °C. Increasing 2-DEG concentration indicates detrapping process. Decreasing 2-DEG concentration indicates trapping process.

or substrate voltage pulse. Charging or discharging of either barrier or buffer electron traps is accompanied by depletion or enhancement of the 2-DEG concentration, respectively. Thus, the evolution of trap occupancy is probed by monitoring the transient response of the 2-DEG concentration. Being a 1-D structure, the electrical field distribution across the heterostructure is better controlled in the gated van der Pauw structure compared with the 2-D transistor structure.

The threshold voltage of our structure is about -5 V, corresponding to a source drain current of 1 mA/mm. In order to probe traps located at different regions of the semiconductor, various stress conditions were applied, as summarized in Table I. The gate current–voltage characteristic of the low leakage van der Pauw test structure can be found in [24].

Fig. 3 presents the 2-DEG concentration transient response, namely, the variation of the 2-DEG concentration compared with equilibrium, following a different gate and substrate voltage stress, at 80 °C. Note the rich behavior, as the shape and polarity of the transients change substantially with gate and substrate voltage. Interestingly, in some cases, an initial increase is followed by a decrease of the 2-DEG concentration, in other cases vice versa, and in some of the cases, a monotonous behavior is recorded. Our data thus indicate that both charging and discharging processes of traps take place during and following the gate pulse. Hence, a dynamic charge redistribution (both during the gate pulse and after the gate pulse) at nonequilibrium conditions determines the 2-DEG transient behavior [13], and thermal statistics, represented by the

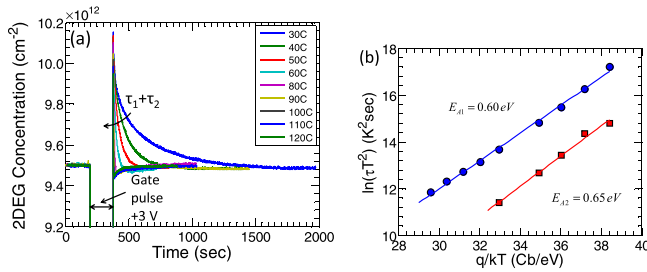


Fig. 4. (a) Gated van der Pauw measurements before and after a 3 min +3 V gate voltage pulse at different temperatures. (b) Arrhenius plot of the main time constants. Two slow transient processes with positive amplitudes were extracted, having activation energies of 0.60 eV for τ_1 and 0.65 eV for τ_2 . The error in the extracted activation energies in all cases is about 0.05 eV, mainly due to temperature fluctuations.

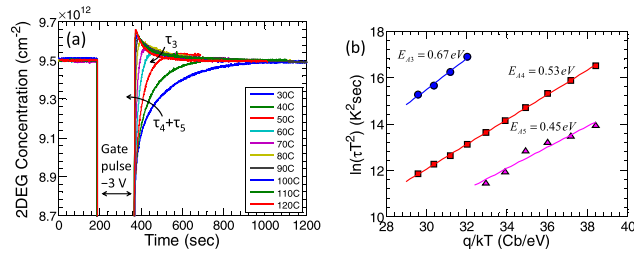


Fig. 5. (a) Gated van der Pauw measurements before and after a 3 min -3 V gate voltage pulse at different temperatures. A transient response of the 2-DEG indicative to either trap charging or discharging was measured. (b) Arrhenius plot of the main time constants that were extracted using a multi exponential fit considering positive and negative preexponential amplitudes. The activation energy of the positive amplitude process τ_3 , indicative of trap charging, is 0.67 eV. The activation energies of the negative amplitude processes, indicative of trap discharging, τ_4 and τ_5 are 0.53 and 0.45 eV.

quasi-Fermi levels, are not a useful concept for understanding the transients. Instead, rate equation must be considered. In other words, both trap filling and trap emptying may take place for the same gate voltage, so that one cannot tell in advance if the net process is filling or emptying of the traps. For example, during a positive gate voltage, AlGaIn barrier traps can either be populated by electrons flowing from the 2-DEG toward the traps, or emptied due to electrons flowing away from the traps to the gate metal. The dominant process depends upon the details of the transport process, and cannot be predicted by the polarity of the applied gate voltage. Hence, the usefulness of the gated van der Pauw method lies in the fact that one can infer whether a depletion process or a filling process of traps prevailed during the gate pulse, simply by monitoring the polarity of the transients after the gate voltage.

The temperature dependence of the 2-DEG concentration transients following a different gate and substrate stress is shown in Figs. 4–7. The dominant time constants of the trap charging and discharging processes were extracted using a multiexponential least-mean-square fitting method, similar to the methodology reported in [12]. Positive and negative preexponential factors represent charging and discharging processes of traps, respectively (an enhancement of the 2-DEG concentration after a gate voltage pulse indicates that trap discharging process took place during the gate pulse, and vice versa). Table II summarizes the extracted activation

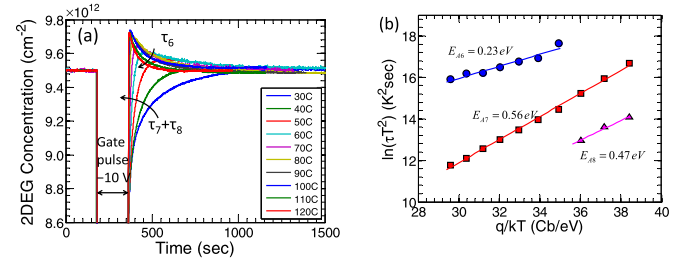


Fig. 6. (a) Gated van der Pauw measurements before and after a 3 min -10 V gate voltage pulse at different temperatures. (b) Arrhenius plot of the main time constants. One positive amplitude process related to detrapping process (τ_6 , with an activation energy of 0.23 eV) and two negative amplitude processes related to a trapping process (τ_7 and τ_8 , with activation energies of 0.56 and 0.47 eV, respectively) were extracted.

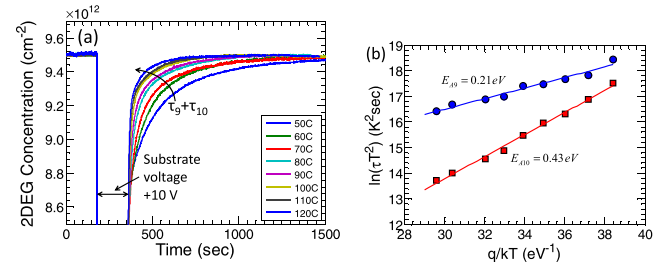


Fig. 7. (a) Gated van der Pauw measurements before and after a 3 min +10 V substrate voltage pulse at different temperatures. (b) Arrhenius plot of the main time constants. Only negative amplitude processes (detrapping) were measured having activation energies of 0.21 eV for τ_9 and 0.43 eV for τ_{10} .

energies and the capture cross sections of the main transient processes at each gate or substrate voltages. Trap charging and discharging processes are shown separately. As further discussed in the following, the extracted activation energies may, in certain cases, be determined by transport rather than by trap energy. In these cases, the value of the “capture cross section” should be translated to some preexponential factor characteristic of the specific transport process. The cross sections we have measured are smaller than reported in many other studies. Since the physical interpretation of the cross section for trapping and detrapping processes as well as for transport processes is not yet well understood, further studies are clearly required.

Following a +3 V gate voltage (that affects only barrier traps), two charging (trapping) processes with similar activation energies of 0.60 and 0.65 eV were found. A similar charging process, with an activation energy of 0.67 eV, was also probed after a semi-ON gate voltage of -3 V. We assign these processes, having similar charging activation energies, to barrier traps (labeled E_{Barrier}). The electrons that charge the traps can flow either from the gate electrode or from the 2-DEG, depending most probably on the spatial location of the traps in the barrier. These two different processes could in principle be distinguished by measuring the polarity of the gate current, which is too low, however, to be detected. We emphasize that since the measured activation energies are characteristic of trapping processes, it conveys no information on the traps energies [25]. Their physical origin is related to the transport process toward the traps, which is currently not understood [24].

TABLE II
SUMMARY OF ACTIVATION ENERGIES AND CAPTURE CROSS SECTION OF THE DIFFERENT TIME CONSTANTS

Process	+3 V Gate voltage		−3 V Gate voltage		−10 V Gate voltage		+10 V Substrate voltage	
	Activation Energy (eV)	Capture cross section, σ (cm ²)	Activation Energy (eV)	Capture cross section, σ (cm ²)	Activation Energy (eV)	Capture cross section, σ (cm ²)	Activation Energy (eV)	Capture cross section, σ (cm ²)
Electron detrapping	—	—	0.53	8.2×10^{-20}	0.56	1.9×10^{-19}	0.21	5.2×10^{-26}
			0.45	4.4×10^{-20}	0.47	9.1×10^{-20}	0.43	6.5×10^{-22}
Electron trapping	0.60	5.7×10^{-19}	0.67	1.5×10^{-19}	0.23	1.9×10^{-25}	—	—
	0.65	3×10^{-17}						

Two activation energies were obtained for the detrapping processes from buffer traps following a -3 V and a -10 V gate voltages. Their values are between 0.53 and 0.56 eV for what we label as $E_{\text{Buffer},1}$, with similar capture cross section in the range of 1×10^{-19} cm², and between 0.45 and 0.47 eV for $E_{\text{Buffer},2}$, with similar capture cross section of about 7×10^{-20} cm². Following a substrate back gating of $+10$ V, a signature of what could be associated with $E_{\text{Buffer},2}$ was recorded, with activation energy of 0.43 eV and smaller capture cross section of 6.5×10^{-22} cm², but not of $E_{\text{Buffer},1}$. We thus conclude that trap $E_{\text{Buffer},2}$ resides closer to the substrate than trap $E_{\text{Buffer},1}$. Since we have only measured detrapping processes, the activation energies may be associated with the trap energies, although transport related activation energies cannot be ruled out as well.

A different electron detrapping event with an activation energy of 0.21 eV (labeled $E_{\text{Buffer},3}$) was found following a $+10$ V substrate back gating. The spatial location of this trap must be closer to the substrate than those traps characterized by $E_{\text{Buffer},1}$ and $E_{\text{Buffer},2}$. We have also identified a trapping process following a -10 V gate voltage with an activation energy of 0.23 eV. A possible interpretation of this process is thermionic assisted tunneling between traps requiring $E_{\text{Buffer},2} - E_{\text{Buffer},3}$, namely, 0.24 eV.

From the above-mentioned analysis, a spatial electron trap energy map emerges, as schematically presented in the energy band diagram of the AlGaIn/GaN HFET structure in Fig. 8. Three barrier trapping processes with similar charging activation energy of about 0.64 eV were identified. In the GaN buffer layer, we probed three different electron traps, a trap with a detrapping activation energy of about 0.55 eV vertically located closer to the 2-DEG channel, a trap vertically located far from the 2-DEG with a detrapping activation energy of 0.21 eV, and a trap with a detrapping activation energy of about 0.45 eV located intermediately between this two traps.

Our results can be correlated with previous works. Meneghesso *et al.* [18] have identified defective levels responsible for a threshold voltage shift in ITO Schottky gate AlGaIn/GaN HFETs with a detrapping activation energy of 0.631 eV and a capture cross section of 4.9×10^{-19} cm². They concluded that the trap is spatially located in the region under the gate, either in the AlGaIn barrier layer or the GaN buffer layer. Having a similar activation energy and a capture cross section as the E_{Barrier} energy levels reported here, our data indicate that this electron trap is most probably located in the AlGaIn barrier layer. Deep level transient spectroscopy measurements [26] have suggested that a GaN buffer trap

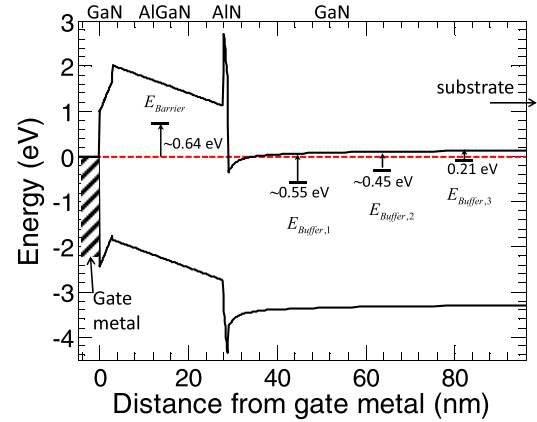


Fig. 8. Simulated energy band diagram of the HFET structure with schematic of the trap locations.

having an activation energy of 0.50–0.60 eV can be due to nitrogen antisites, while a GaN buffer trap having an activation energy of 0.22 eV can be attributed to nitrogen vacancies. Caesar *et al.* [27] have performed drain current trapping transients, and have detected a GaN buffer trap with activation energy of 0.44 eV. They concluded that this trap might be due to C/O/H impurities, possibly in nitrogen substitution position. Obviously, more studies are required to understand the nature of traps in AlGaIn/GaN HFET structures.

IV. CONCLUSION

In this paper, we studied electron trapping effects in AlGaIn/GaN HFET structures using the gated van der Pauw method. We have demonstrated how different trap locations can be identified and discriminated by changing gate or substrate voltage. We identified several traps in our GaN-based heterostructure on Si substrate, and one trapping activation energy of about 0.64 eV was associated with trapping processes in the AlGaIn barrier layer. We also identified three traps located in the GaN buffer layer with detrapping activation energies of about 0.55, 0.45, and 0.21 eV. This method thus enables easier identification of traps in GaN-based HFET heterostructure and thereby contributing to the physical understanding of electron trapping effects in GaN-based HFET technology.

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